



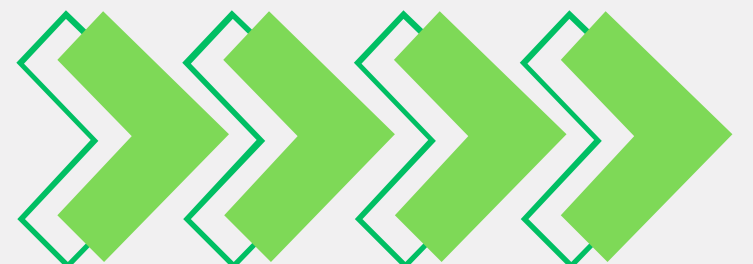
Written in English for clarity — some technical terms don't translate naturally yet

Embedding Legal Data: The Key to Measuring Context and Meaning

Introducing the Indonesian Legal RAG Dataset—The first corpus to provide pre-computed vectors for building AI that understands the spirit of the law, not just keywords.



Afza Azzindani



The Next Step After Structured Reasoning

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- **The Context (My R&D):** My previous work successfully fine-tuned an LLM for Indonesian law (using GRPO), achieving excellent structured reasoning and logical flow.
- **The New Problem:** While that model could reason, it suffered from low specific regulation accuracy—it couldn't reliably cite the exact article because its internal knowledge was static and imprecise.
- **The Solution Imperative:** To solve this, we must build a Retrieval-Augmented Generation (RAG) system. This requires a specialized dataset with pre-computed, high-quality vectors to guarantee the LLM retrieves and quotes the correct article every time.
- **Goal:** This dataset is the backbone for fixing the "citation problem" and ensuring our Legal AI is finally trustworthy and citable.

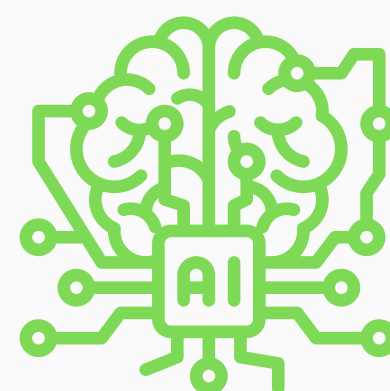


Building a Trustworthy AI (The RAG System)

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To fix the LLM's citation problem, we must integrate a system known as Retrieval-Augmented Generation (RAG).

- **RAG as a Librarian:** Imagine the LLM has a "knowledge cutoff." RAG acts as a highly efficient librarian, allowing the LLM to access live, external documents in real-time.
- **How it Works:** When a user asks a legal question, the RAG system first performs a deep search across its external knowledge base (our 5,000 legal documents).
- **Data Feeding:** The system finds the most relevant articles or sections and then feeds that retrieved text directly to the LLM.
- **The Result:** The LLM is forced to answer the user's question using only the verified context from the legal documents it was just handed, ensuring the response is grounded, current, and citable.



The Challenge: While RAG's concept is simple, building the suitable search engine—the retrieval mechanism—from scratch is the most time-consuming, resource-intensive, and failure-prone part of the entire project.



Keyword vs. Context

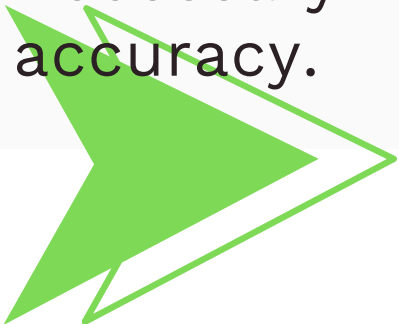
Building a reliable legal RAG requires balancing two distinct search methods, as a single method always misses key information:

	Keyword Search (TF-IDF Vector)	Context Search (Dense Embedding)
Mechanism (How it Works)	Statistical measure of term frequency (sparse vectors)	Numerical representation of the meaning and semantic relationships
Advantage (What it Finds)	High Precision: Finds exact words, article numbers, and legal citations	High Recall: Finds documents related by concept, even if exact words are not used
Limitation (What it Misses)	Fails if wording is slightly different; misses documents with the same meaning but different vocabulary	Can miss the exact legal citation/number; struggles with technical specificity if the context window is too broad

Example:

- **Query:** "hak-hak pekerja yang di-PHK" (rights of workers who are terminated)
- **Context Search (Embeddings):** Finds documents using the term "pemutusan hubungan kerja" (termination of employment).
- **Keyword Search (TF-IDF):** Finds documents with the exact term "di-PHK" and the specific UU Nomor 13 Tahun 2003.

Instead of using one method, I **use both—dense embeddings (context) and TF-IDF (keyword)**—to build a RAG dataset suitable for Indonesian legal documentation, because of the necessary balance between conceptual understanding and citable accuracy.



The Embedded Legal Foundation



This stage marks the completion of the embedding phase — transforming raw legal text into structured, retrievable intelligence

Dataset Summary:

- **Parsed Source:** ~10,000 Indonesian regulations
- **Output Scale:** ~100,000 article-level rows (atomic article level)
- **Embedding Model:** [Qwen3-Embedding-0.6B](#) (100+ languages supported, including Bahasa Indonesia)
- **Dense Vector Size:** 1,024-dimensional semantic embeddings
- **Sparse Vector Size:** 5,000-dimensional TF-IDF keyword vectors
- **Data Schema** (What You Get):
 1. **Metadata:** Regulation Type, Enacting Body, Year, Effective Date.
 2. **Content:** Full document text (for context feed).
 3. **Vectors:** The embedding list and the TF-IDF list.
- **Repository:** [Azzindani/ID REG MD Embed](#)



Immediate Utility and Future R&D Roadmap



This dataset is your engine for immediate and advanced Legal AI development:

- **Key Utility:**

1. **Semantic Search:** Use the 1024-dim embeddings to find documents based on conceptual meaning.
2. **Keyword Search:** Use the TF-IDF vectors to retrieve results based on precise citations and exact terms.
3. **Hybrid Search:** Combine both vectors to build a robust, high-precision retrieval system.
4. **Clustering & Analysis:** Perform large-scale legal document clustering and comparative analysis.

- **Future Roadmap:**

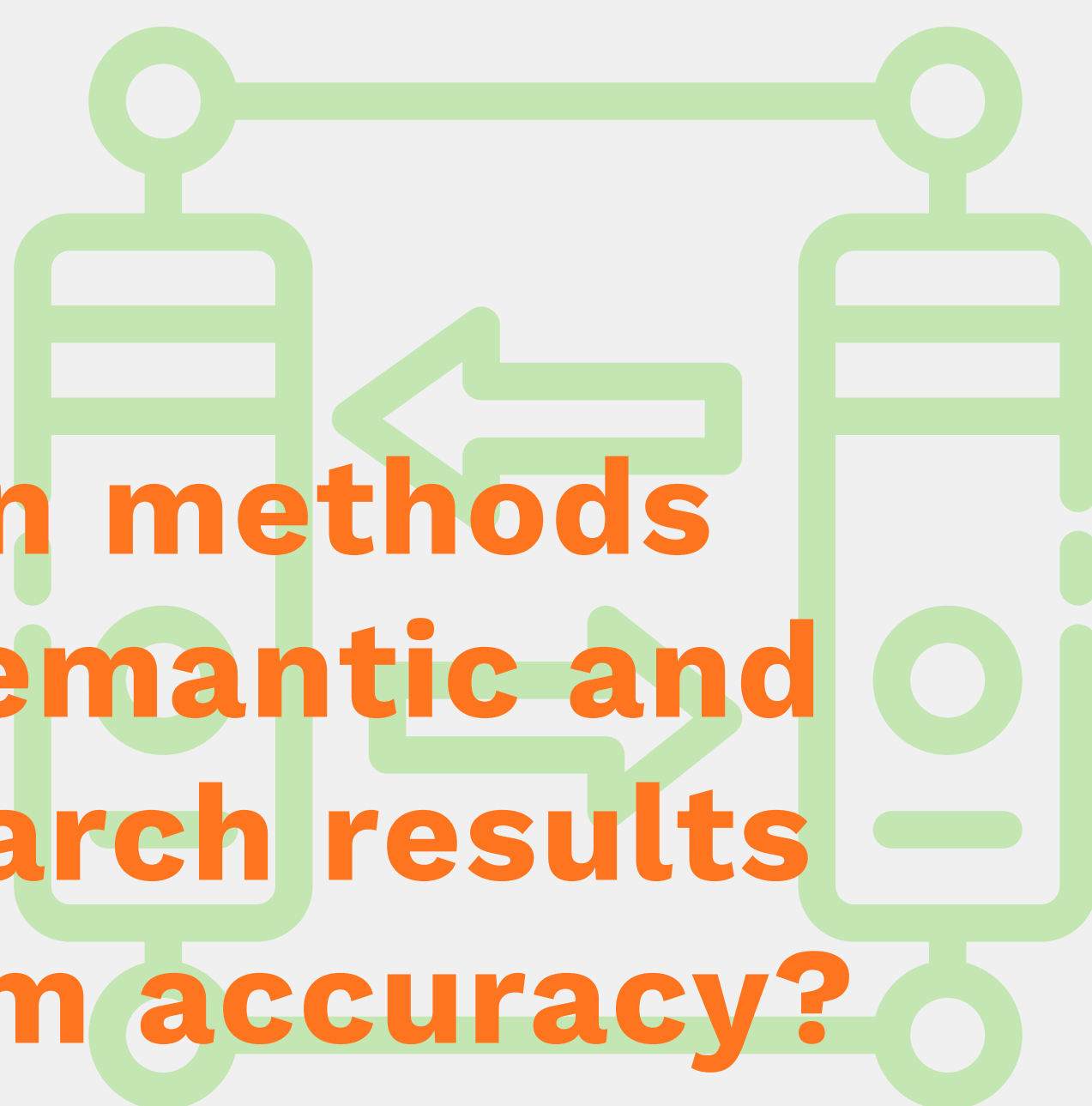
1. **Knowledge Graph Integration:** Build a Legal Knowledge Graph on top of this structured data to map complex relationships between specific articles and related regulations.
2. **Targeted Re-Embedding:** Identify documents where the Qwen3 embedding model is known to perform poorly and re-embed only those documents using a specialized legal language model.

- **Acknowledgements:**

1. [Qwen team](#)



The data is structured, indexed, and ready to solve complex compliance issues.



Interested in methods for fusing semantic and keyword search results for maximum accuracy?

Share your insights and help us validate this crucial R&D asset.